

Online Shop (Nowruz Project)

1. Introduction

In this project, we made an online shop called **Shop 059**, similar to websites like Amazon and Digikala.

There are three different types of accounts in the project: **Admin**, **User**, and **Seller**. Each type has different access and responsibilities.

There is also a **Product** superclass and different product classes that inherit from it: Book, Bag, Cloth, Headphones, Mobile, Painting, Rug, Shoes, SportEquipment, Toy, and Watch.

I also added some example accounts that can be used to enter the program and test the different menus.

- User: firstuser / pass1234
- User: heli / heli7496
- Seller: Goody / 1212pass
- Admin: mainAdmin / 1973admin

2. Product IDs and Requests

I used IDs for products so that each product can be identified separately.

Each product ID has three or four numbers. The first number represents the product category. Since there are 11 categories, some IDs need four numbers. The second number represents the seller. For example, the seller **Goody** has the number 1. The last part represents the product number in that category. For example, if it is 2, it means it is the second product of that category.

I also used a file to make a connection between admins and users/sellers. When an action needs confirmation from an admin, a request is written to the file. When an admin logs into the program, they can see the requests and confirm them. After the request is confirmed, the user or seller can continue what they wanted to do.

Some of the requests are getting authorization for a seller, confirming an order, and adding funds to a wallet.

I made some of the shop attributes `final` because I did not want them to be changed.

3. Menus

3.1 mainMenu()

In the main menu, the predefined objects are added to their related ArrayLists.

There are four ways to enter the program: login as a user, login as a seller, login as an admin, or create a new account as a user or seller. Admins can only be added by another admin.

If you already have an account, you enter the requested information, such as username/company name and password. The login method in the Shop class checks the information, and if it is correct, you can enter the related menu.

If you do not have an account, you can enter the required information and create a new account.

When choosing a username or company name, the program checks if it has already been used and shows an error if it is not available.

3.2 userMenu()

- Search for a product by its name.
- See products in different categories.
- View the shopping cart and the products inside it.
- Remove products from the shopping cart.
- Finalize the shopping cart.
- Add funds to the wallet by sending a request.
- See previous orders and purchased products.
- Edit personal information such as username, password, phone number, and email address.
- See information about the shop, such as its name, support phone number, and support email.

Before a user can finalize the shopping cart, the total price is compared with the amount in their wallet. If the total price is more than the wallet balance, the user cannot finalize the cart.

When a user wants to finalize an order, a request is added to the file. The order can then be confirmed by an admin. After sending the request, the user can check the status of their order from the order status option.

3.3 sellerMenu()

- Add a new product by entering the required information for its category.
- View the products available in their account.
- See the amount of money in their wallet.

3.4 adminMenu()

- Add new admins by entering the required information.
- View users' information.
- View requests made by users and sellers.
- Confirm requests.

After an admin confirms a request, that request is removed from the file.

4. Problems and How I Fixed Them

- I made the attributes of the Admin class protected so that they could also be used in the User class.
- I used an ArrayList for all products and another ArrayList using Object, so different product classes could be stored in the same list.
- I made the predefined objects in the Main class static so I could use them in different scopes.

5. Extra Parts

I used regular expressions (regex) for three parts of the program: password validation, email validation, and phone number validation. If a user, admin, or seller enters an invalid value, the program shows an error and explains what is wrong.

I also used HashMap to find objects using a String key. For users and admins, the username is used as the key, and for sellers, the company name is used.

There are three HashMaps, one for users, one for admins, and one for sellers. This makes finding these objects easier.